TEAM
ACE OF SPADES

CASE STUDY ANALYSIS

ANALYSIS OF BANK **COSTUMER CHURN** AND TO PROVIDE SOLUTIONS TO THE PROBLEM STATEMENTS SO AS TO PROVIDE INNOVATIVE DATA DRIVEN STRATEGIES TO RETAIN HIGH VALUE COSTUMERS AND IMPROVE OVERALL COSTUMER EXPERIENCE.

CSV MATRIX 2.0

THE TEAM

ADITYA RAJ TEWARY (2023UGMM006)
FILZA AHSAN (2023UGMM028)
AMIT KUMAR MAHATO (2023UGMM004)

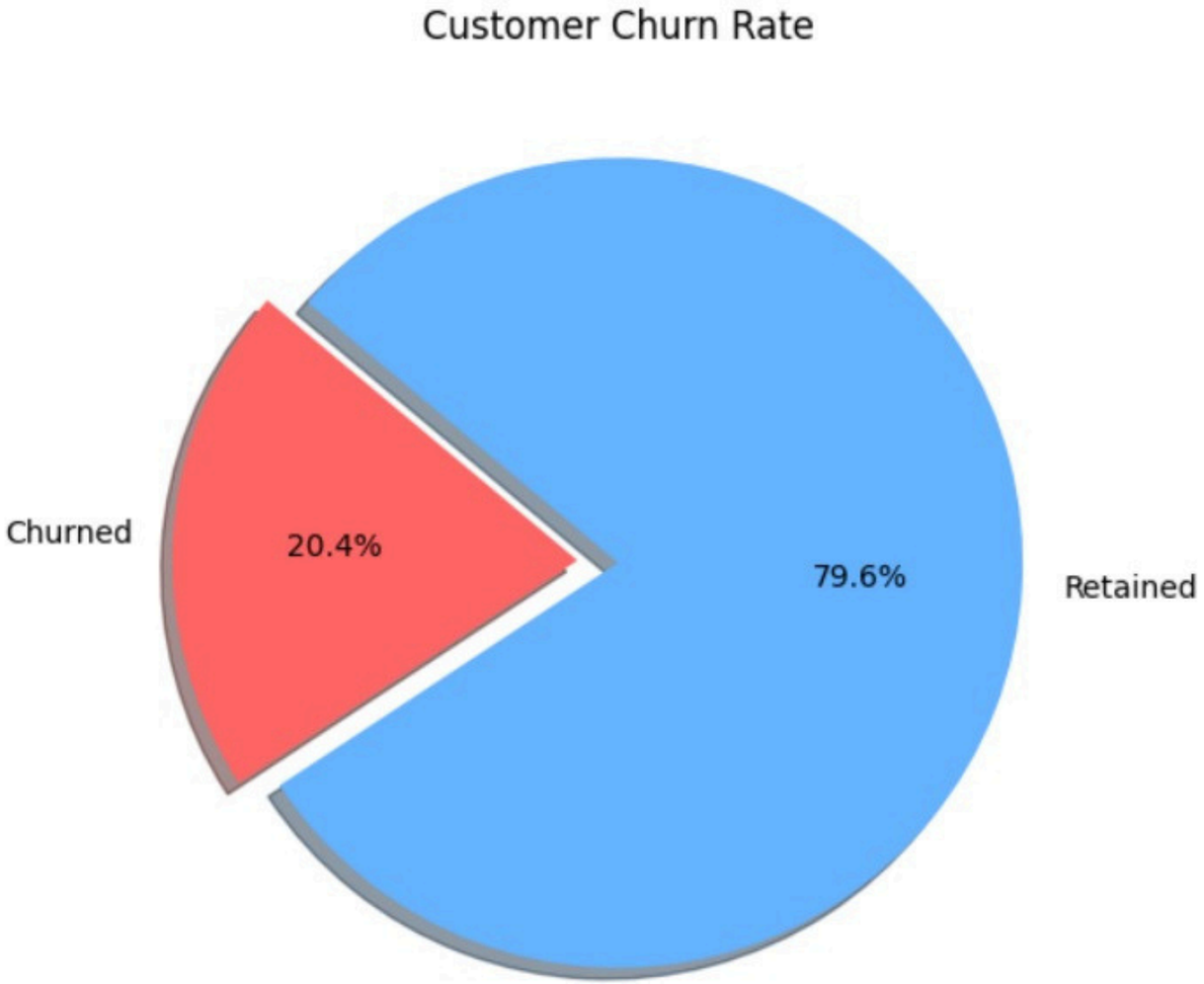
TOOLS USED

PYTHON LIBRARIES: NUMPY, MATPLOTLIB, PANDAS, SEABORN
GOOGLE COLAB, MS EXCEL, GOOGLE SHEETS

THE PROBLEM STATEMENT

ABC Bank, a leading financial institution, is facing a significant challenge with customer retention. Over the past few years, an increasing number of customers have been closing their accounts or shifting to competitor banks. To address this issue, the bank's data team has collected detailed customer information, including demographic data, account details, transaction history, and customer behavior patterns. Despite their efforts, the churn rate continues to rise, and ABC Bank is seeking innovative data-driven strategies to retain high-value customers and improve overall customer satisfaction.

Q1 Calculate the overall churn rate of the bank. Also determine if there is a relationship between Tenure(Categorize customers by their tenure) and their likelihood to churn.



FORMULA FOR CHURN RATE

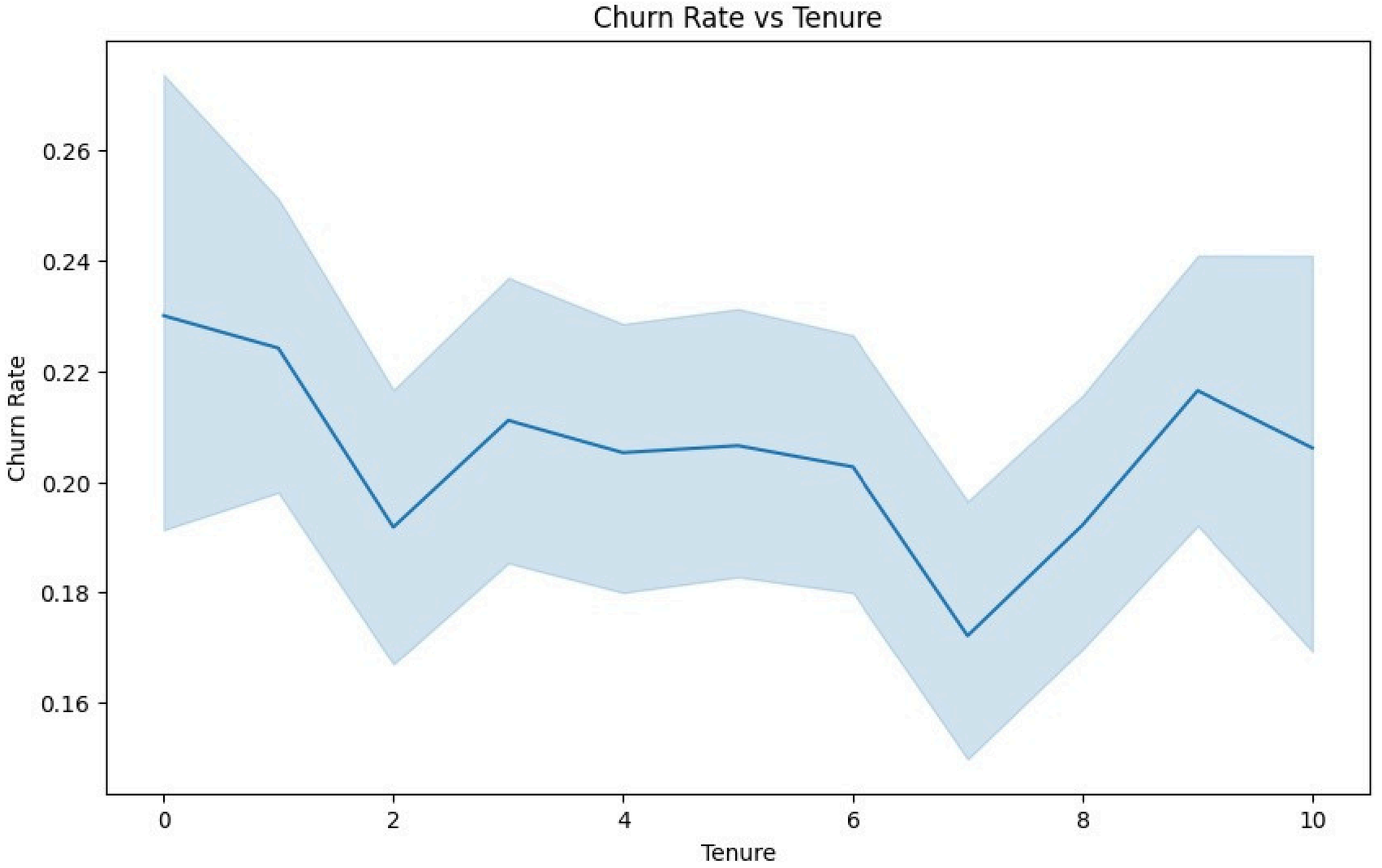
$$\text{CHURN RATE} = \frac{\text{COSTUMERS EXITED}}{\text{TOTAL COSTUMERS}} \times 100$$

METHOD : Using **COUNT()** function

- The total number of costumers who exited alongwith the total number of costumers was counted.
- Then by putting the data in the formula we get the value of churn rate to be **20.37%**
- Thus we then use the final value obtained for churn rate and retained rate to plot the following pie chart.

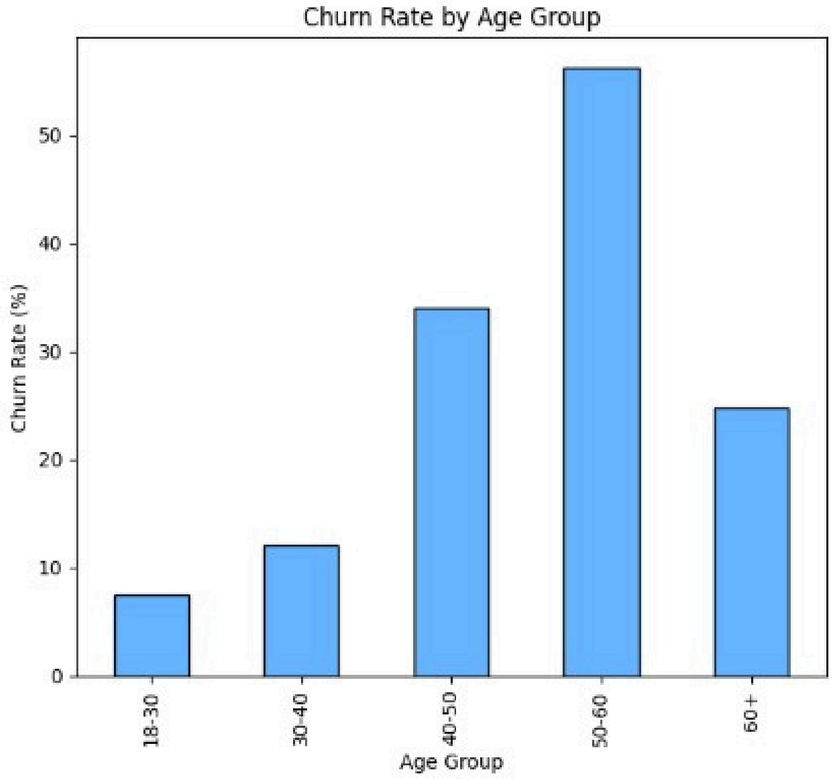
Q1

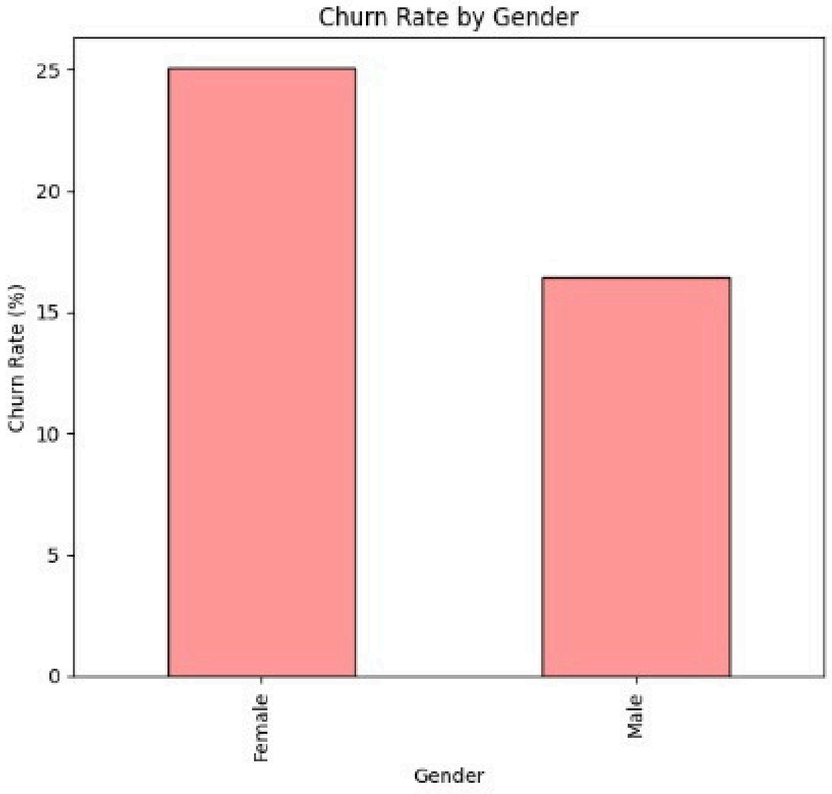
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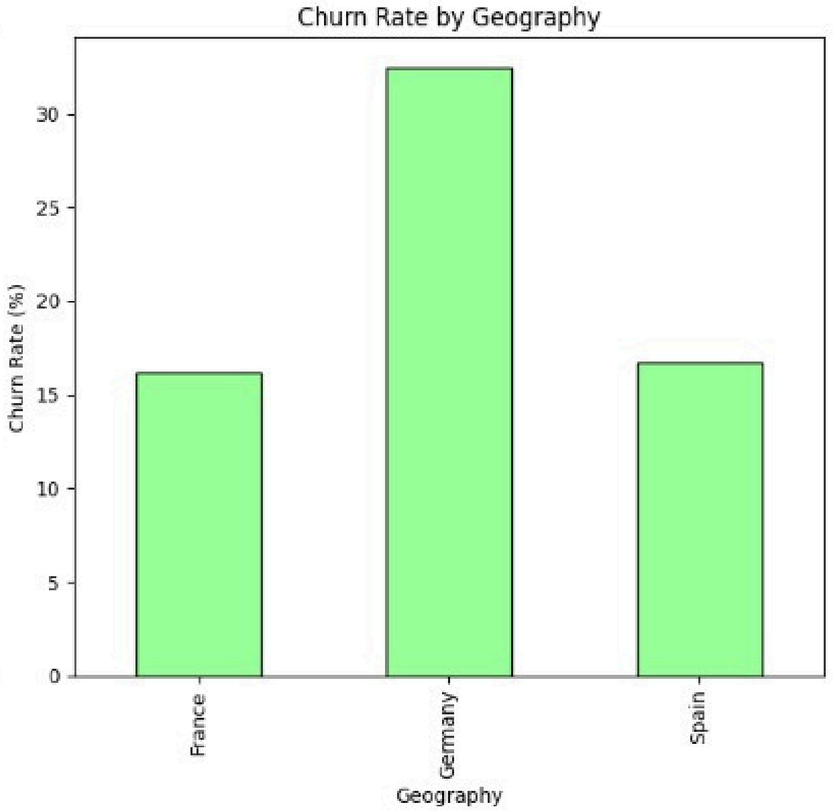


- As such there is no linear relationship between the Churn rate and Tenure.
- But we can observe that the new costumers and the costumers with highest tenure are more likely to churn out.
- The reason might be that they are unaware of the newer services of the bank. The awareness of new services schemes might lead retention.

Q2 Analyze churn based on customer demographics (age, gender, and geography). Which demographic groups have the highest churn rate? Create a bar chart to display this.

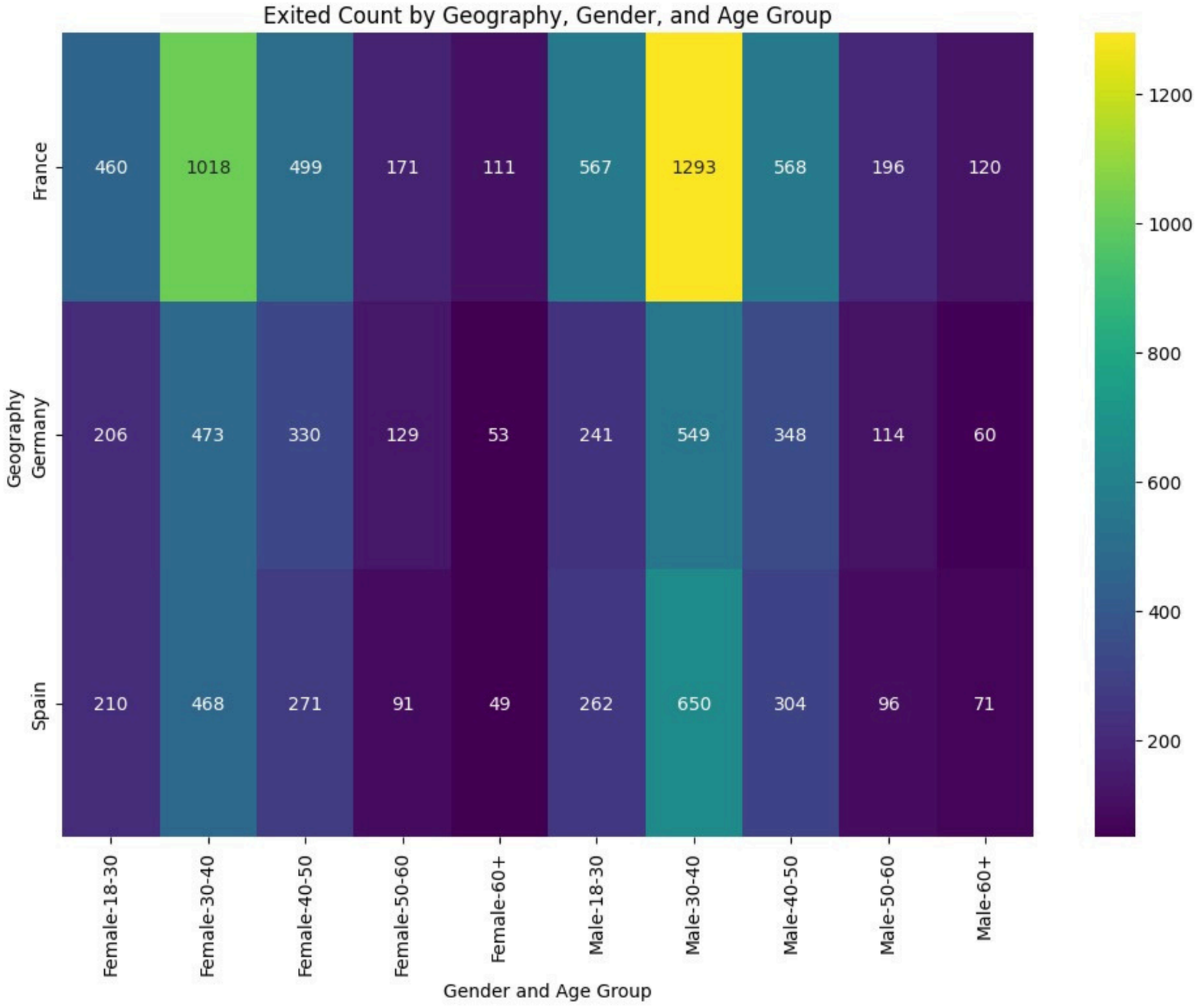






- When solely observing age vs churn rate, customers belonging to the **50-60 age** bracket are most likely to churn.
- When observing the data according to gender, **female customers** are more likely to churn
- When observing the geographical demographic, we see that customers from **Germany** are most likely to churn.

Q2

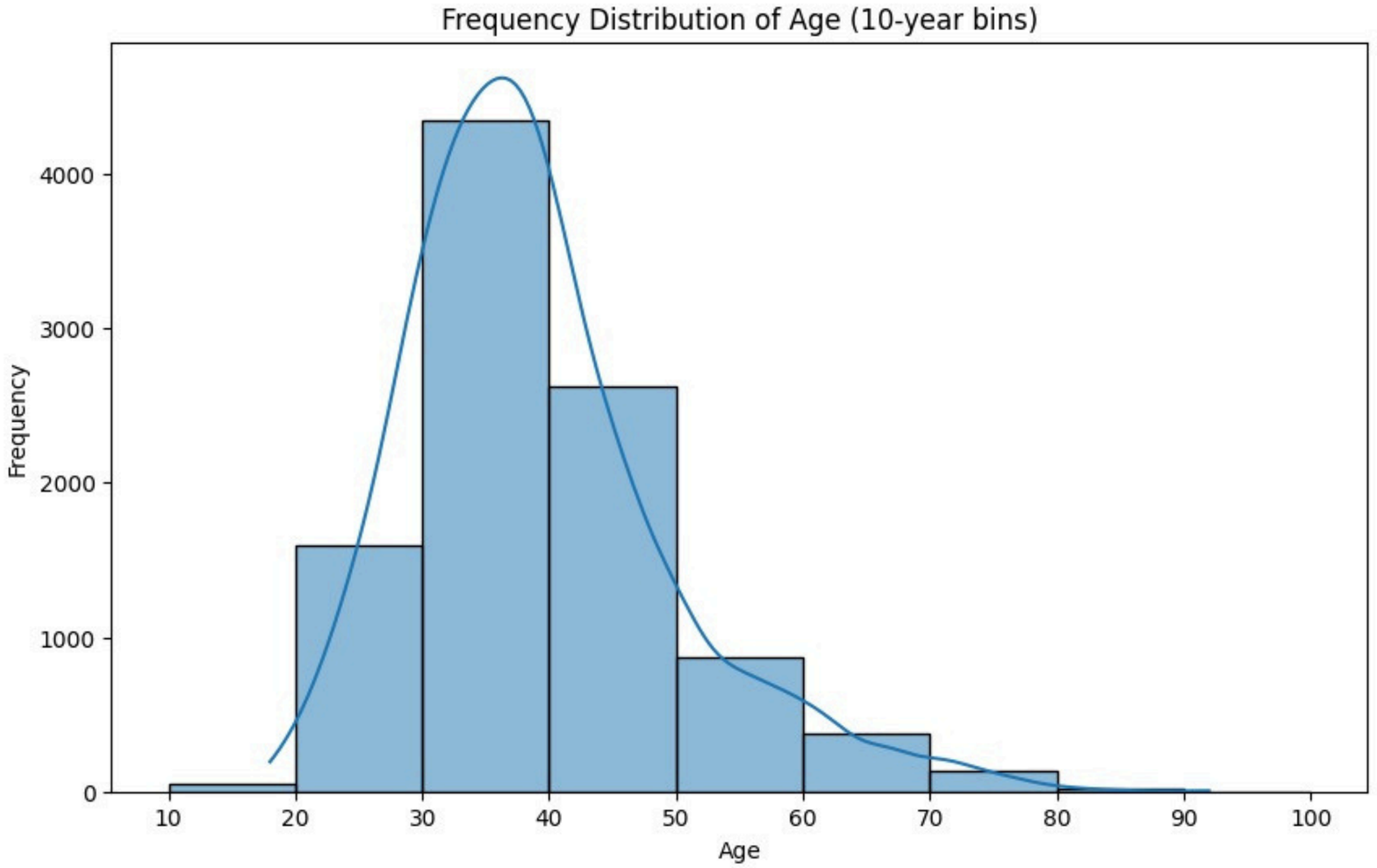


- To gain deeper insights into the combined effects of the three variables, a heatmap visualization was utilized. The analysis reveals that the highest churn rate is observed among **French male customers within the 30-40 age bracket.**

Q3 **You are an consultant in O11 Solutions, and currently your client who are in fintech industry wants to expand their products in France. They want to target customers having credit score> 500, estimated salary > 120000 and has credit card with them. Find the desired results for them. Also, suppose your client is offering fintech services, describe brief framework they should follow to enter the markets in France.**

- As a consultant and a data analyst I have created a new data set obtained after filtering out the costumer data according to the conditions to be met for the required target costumers. The dataset can be accessed below.
- Link to the dataset with required filters.
- Various Python Operators such as AND, > and == were used accordingly.
- Further for concluding with a market expansion framework in France, frequency distribution of costumers based on Age, Tenure and Credit Score was done along with some market analysis.

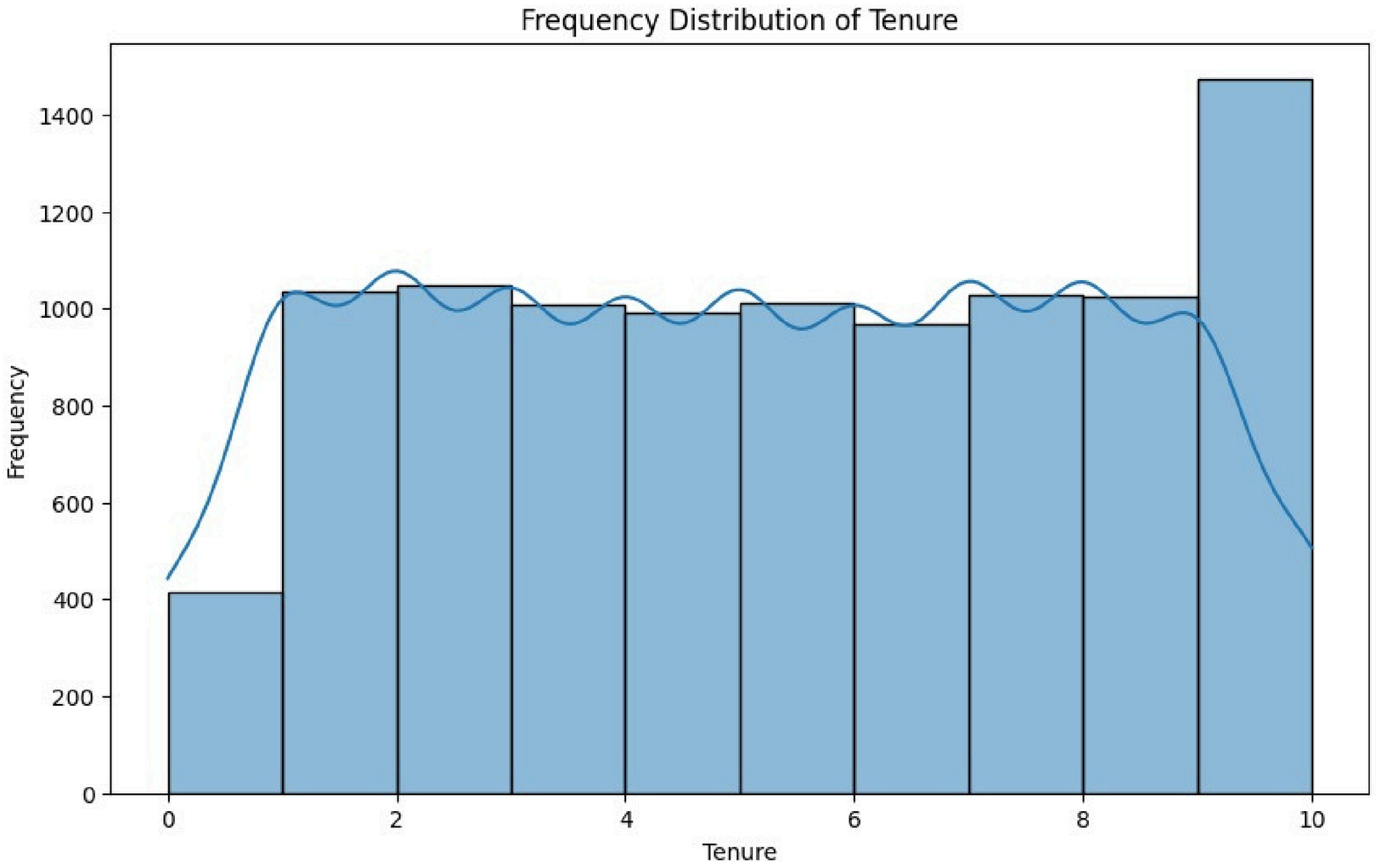
Q3



Frequency Distribution of Age

- When considered the distribution based on age, the highest number of users are found in the **30-40** age bracket. It would be easier to market services to this bracket before others to ensure retention in the market.

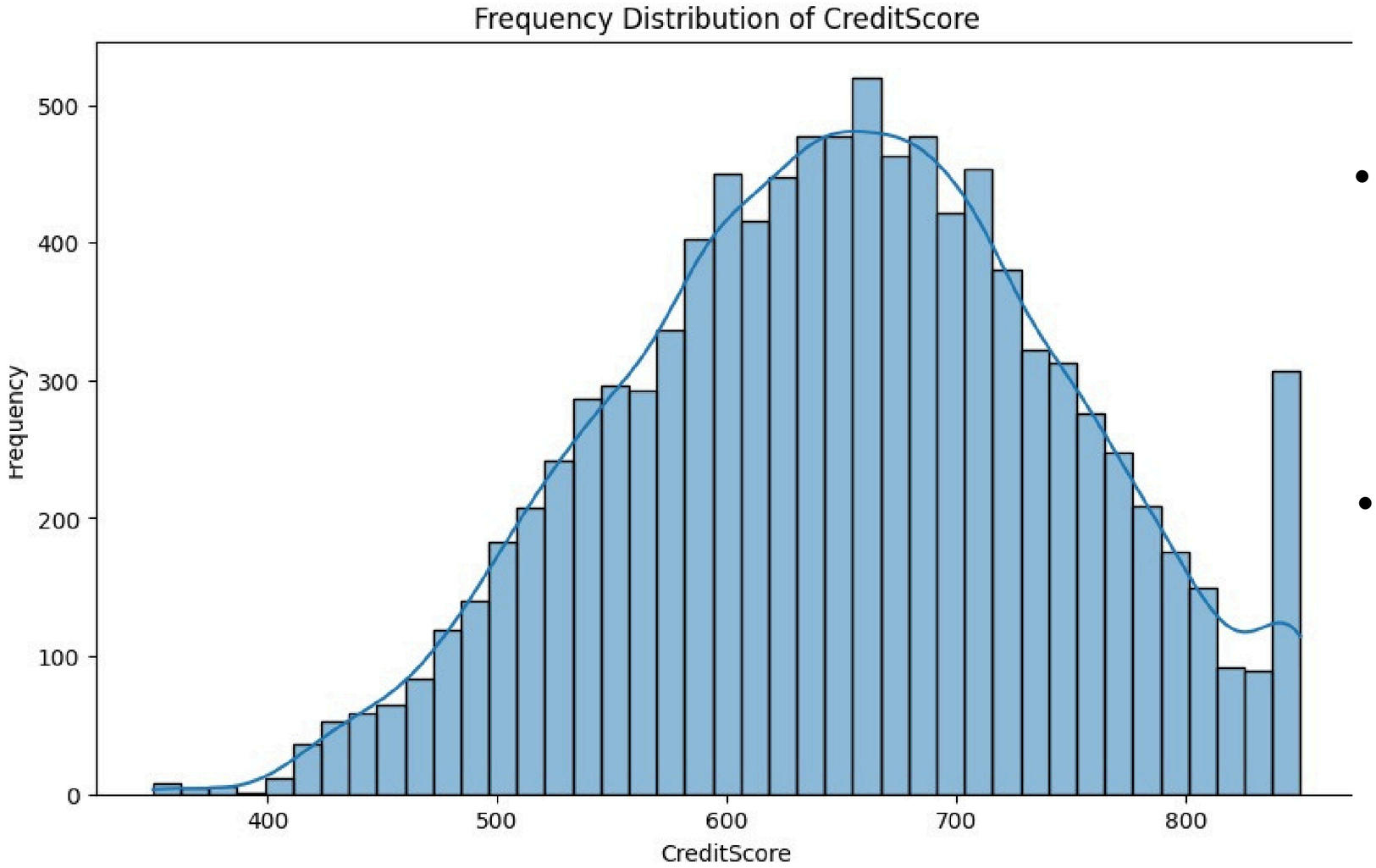
Q3



Frequency distribution of Tenure

- When the distribution is done based on tenure, almost all tenure groups included equal number of costumers hence we can conclude that all tenure groups be equally targeted.

Q3



Frequency Distribution of Credit Score

- When considered the distribution based on Credit Score, the target group should belong to the major section of **600-700** as it involves most costumers.

- The spike at the 800+ range could imply a subset of individuals with excellent credit or could represent specific credit policies that cap or promote high scores

Q3

Here is a framework for Market Entry:

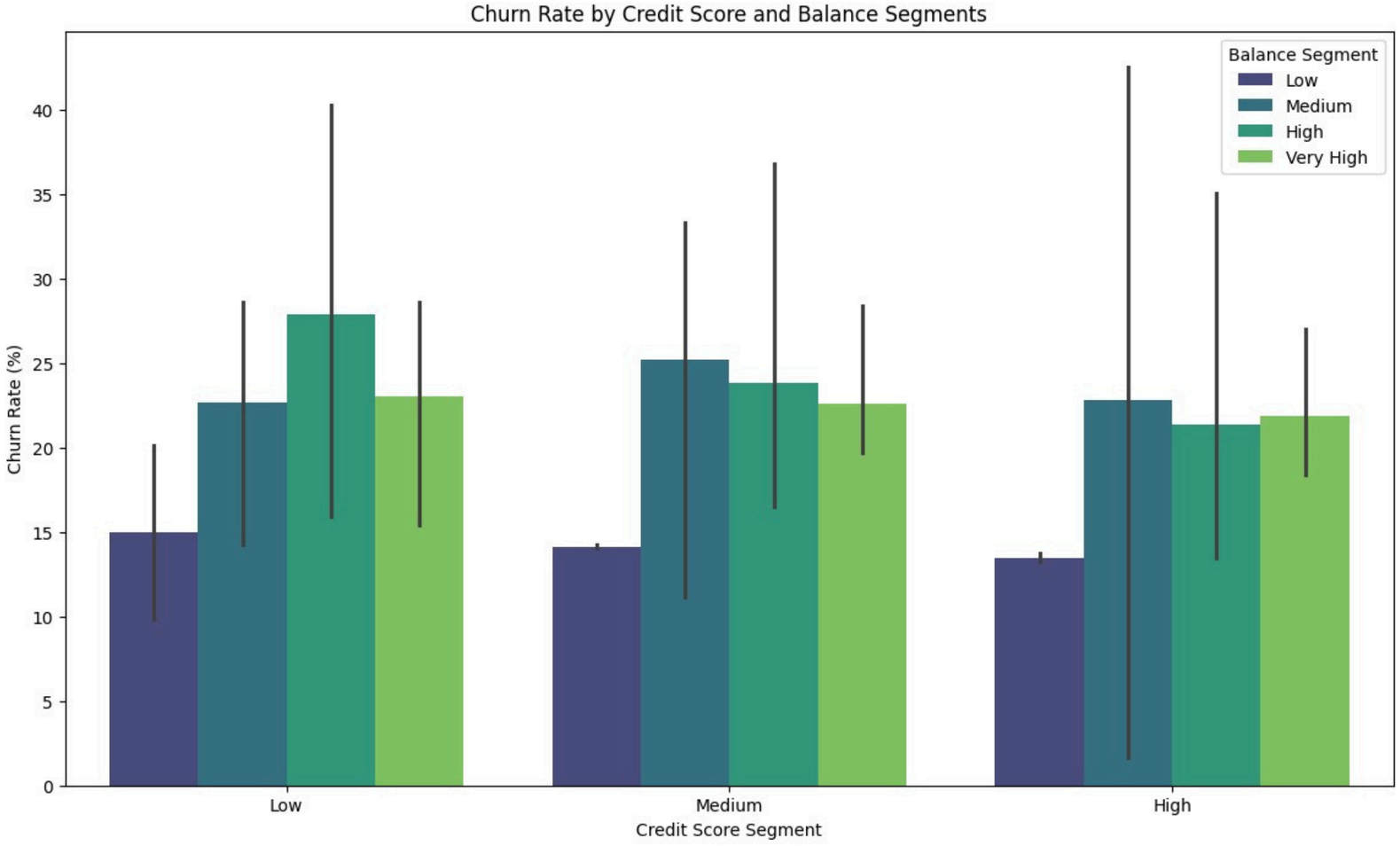
- Market Research: Apart from the research that has been performed using Data Analysis in this case study, ground level market research and customer preferences should be understood considering more criterion than those mentioned in the data.
- Legal Regulations and Compliances: The client should work with legal experts in France to understand the consequences and requirements of entering into the country's Fintech space and obtaining required licenses.
- Marketing: Customer Acquisition Cost has to be identified within a reasonable budget after segregating the target customer base.
- Social Media outreach: Reaching out to local influencers and printed advertisement helps in propagation of the brand. Social media can be used to spread the word for stalls of the company in various conclaves that could help in organic growth of the brand

Q3

Here is a framework for Market Entry:

- Customer outreach: Local customer outreach should be set up in regional languages.
- Security: An encrypted payment and banking service should be set up and demonstrated to customers to win their trust in the service and convince them to opt for this particular service.
- User feedback: User feedback should be taken seriously to optimise various functions of the service.
- Monitoring: KPIs should be continuously monitored for regulation and optimisation of the service.

Q4 Create customer segments based on Credit Score and Balance ranges then summarize their churn by each segment. Which credit score segment has the highest average account balance, and how does this compare across different geographies?



- X-axis (Credit Score Segment): Divided into three segments — Low, Medium, and High credit scores.
- Y-axis (Churn Rate): Shows churn rate as a percentage.
- Balance Segments (Color-coded): Each credit score segment is further divided by balance categories

Balance ranges

LOW	0-25000
MEDIUM	25000-75000
HIGH	75000-150000
VERY HIGH	150000-200000

Q4

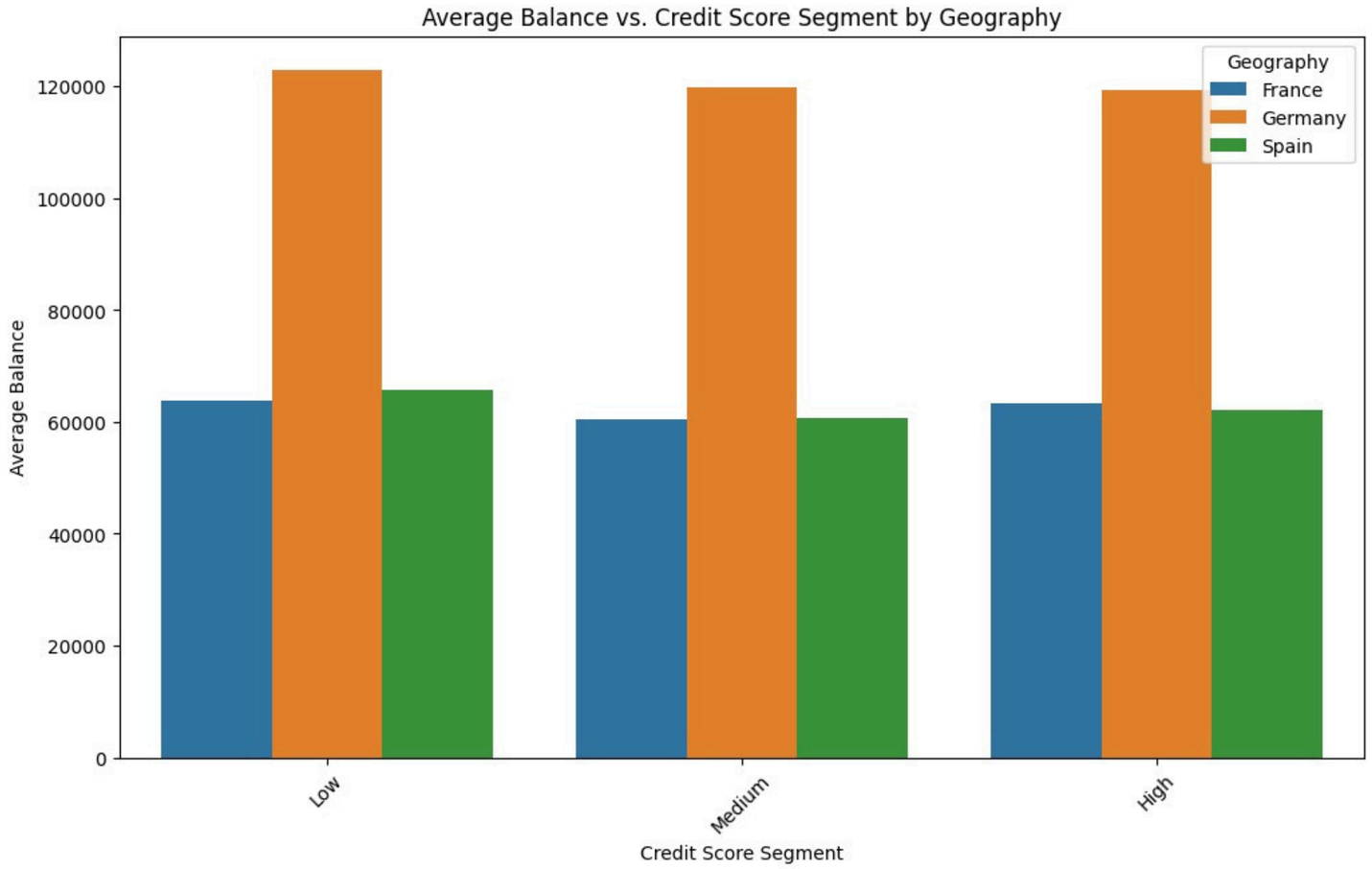
Analysis:-

- Churn rates decrease as credit scores improve.
- Within each credit score segment, Low balance customers consistently show the lowest churn rates.
- Higher balances, particularly in the Low Credit Score group, are linked to higher churn rates.
- **Low Credit Score with High Balance** shows the highest churn, suggesting financial stress or risk factors.
- The error bars vary in length across segments, indicating variability in churn rates within each group. Longer error bars suggest greater variability or uncertainty in the churn rate measurements for that group, especially visible in the Medium and High balance segments.

Q4

Analysis:-

The credit score segment with the highest average account balance is: Low



Credit Score

Low credit Score	**300-500**
Medium Credit Score	**500-650**
High Credit Score	**650-850**

Q5 Identify 'At-Risk" customers based on the criteria: Credit Score < 600, Age > 50, Balance > 100,000, IsActiveMember = 0, NumOfProducts ≤ 1 & Exited =0. How many customer are At-Risk of attrition ?

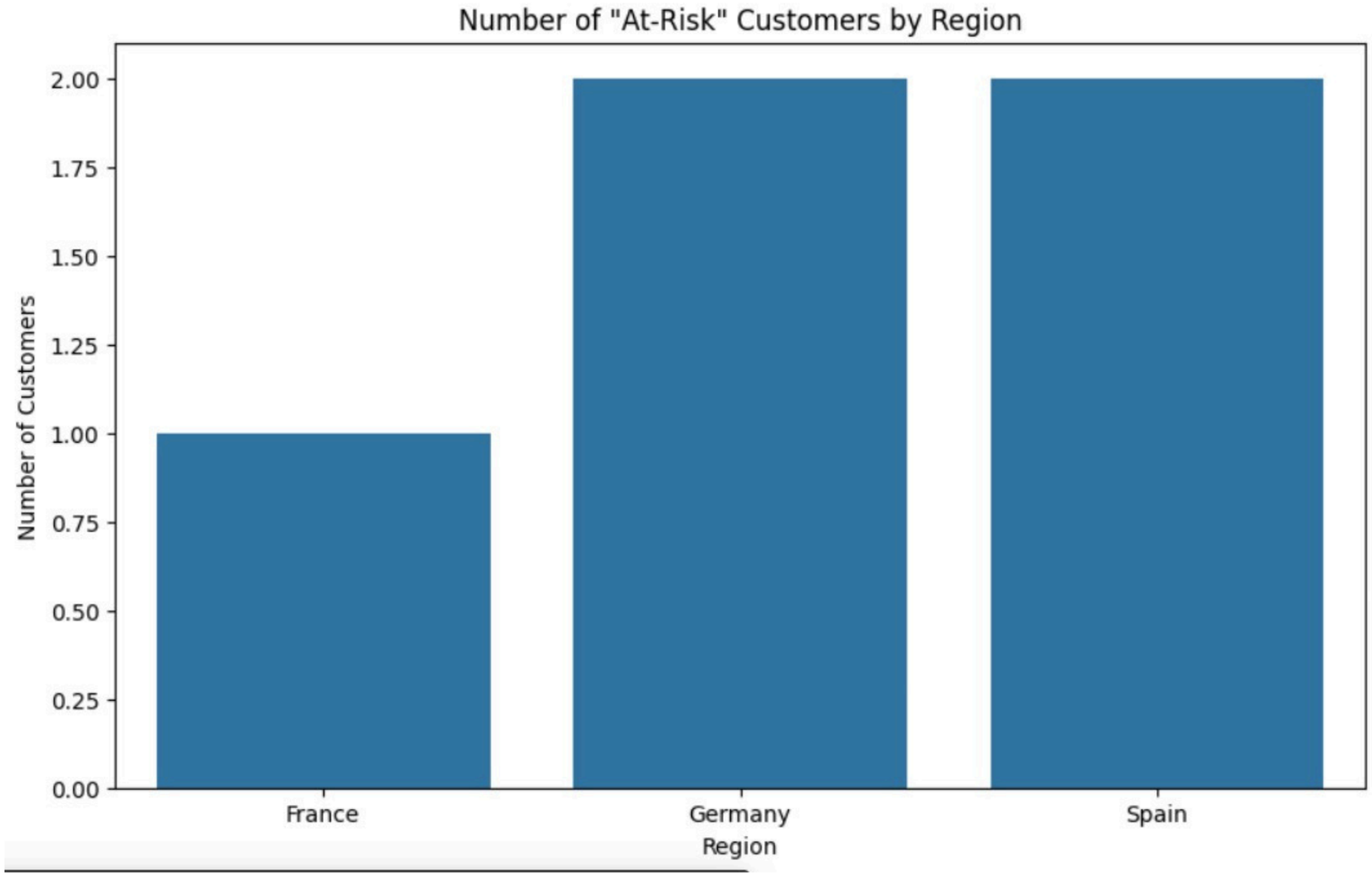
We created a new dataset of "At-Risk" customers as follows after applying the required filters in the various categories/fields.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
tomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProd	HasCrCard	IsActiveMember	EstimatedSalary	Exited	CreditScoreSegment	BalanceSegment	CLV	CreditScoreGroup	AgeGro
.5785406	Watts	446	France	Female	51	4	105056.1	1	0	0	70613.52	0	Low	High	282454.0	[400, 500)	50-59
.5785576	Mayrhofer	434	Germany	Male	71	9	119496.8	1	1	0	125848.88	0	Low	High	1132639.	[400, 500)	60+
.5623314	Tucker	506	Germany	Female	59	3	190353.0	1	1	0	78365.75	0	Medium	High	235097.2	[500, 600)	50-59
.5673995	Tu	516	Spain	Female	65	9	102541.1	1	1	0	181490.42	0	Medium	High	1633413.	[500, 600)	60+
.5693690	Iweobiegbunam	574	Spain	Male	52	7	115532.5	1	1	0	196257.67	0	Medium	High	1373803.	[500, 600)	50-59

A total of **5** costumers were found to be at-risk of Attrition.

Q6

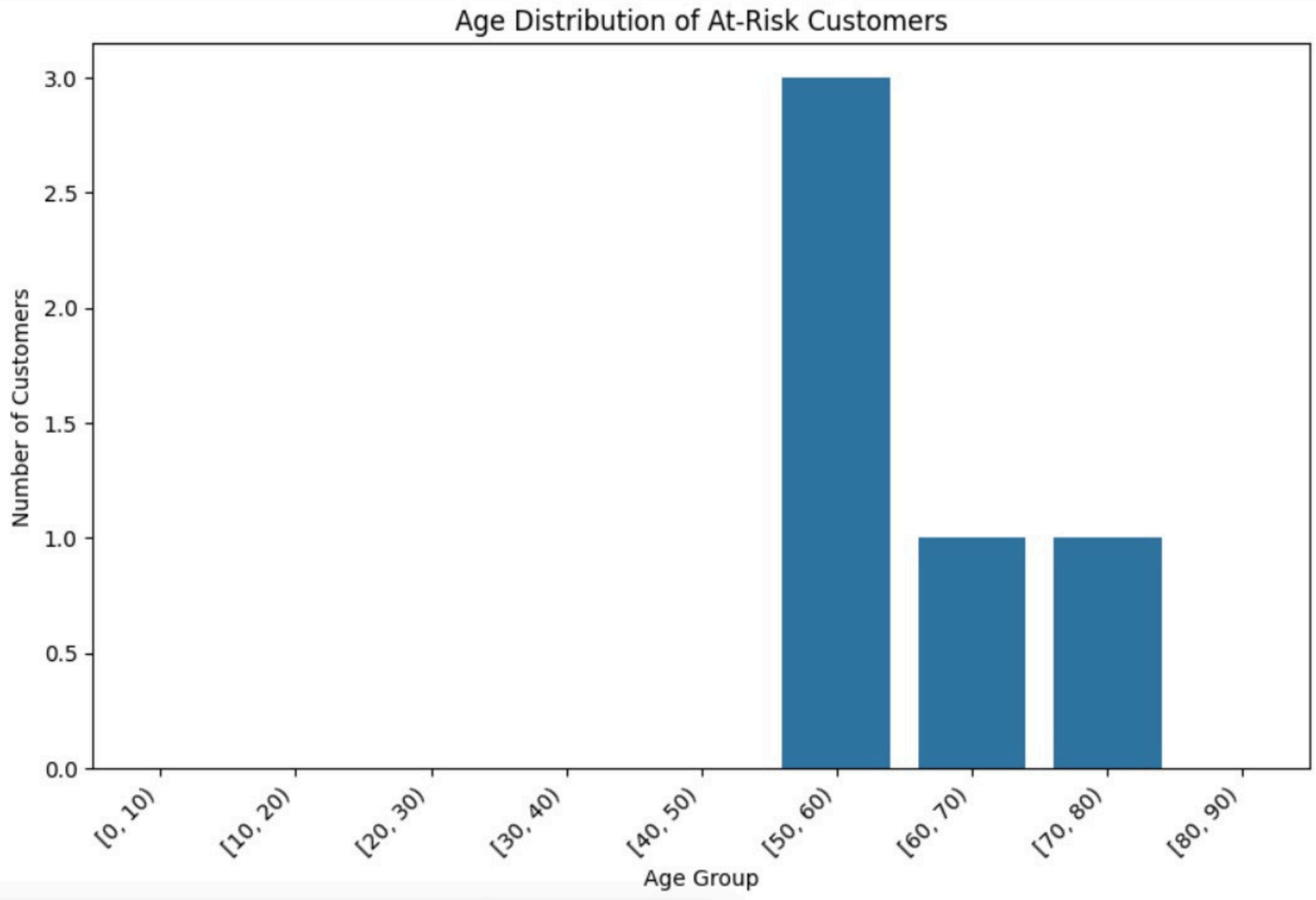
From Q5, Analyze through a Chart a.) Customer in which specific region is more likely to be "At-Risk" b.) Customer of which specific age group is more likely to be "At-Risk" c.) How many of the "At-Risk" Customer doesnot have a credit card ?



- As we can see from the data given that costumers from the **Germany** and **Spain** region are equally At-Risk

Q6

From Q5, Analyze through a Chart a.) Customer in which specific region is more likely to be "At-Risk" b.) Customer of which specific age group is more likely to be "At-Risk" c.) How many of the "At-Risk" Customer doesnot have a credit card ?



- After the graphical analysis of the reduced data set, the consumers At-risk belong majorly to the age group of **50-60 years.**

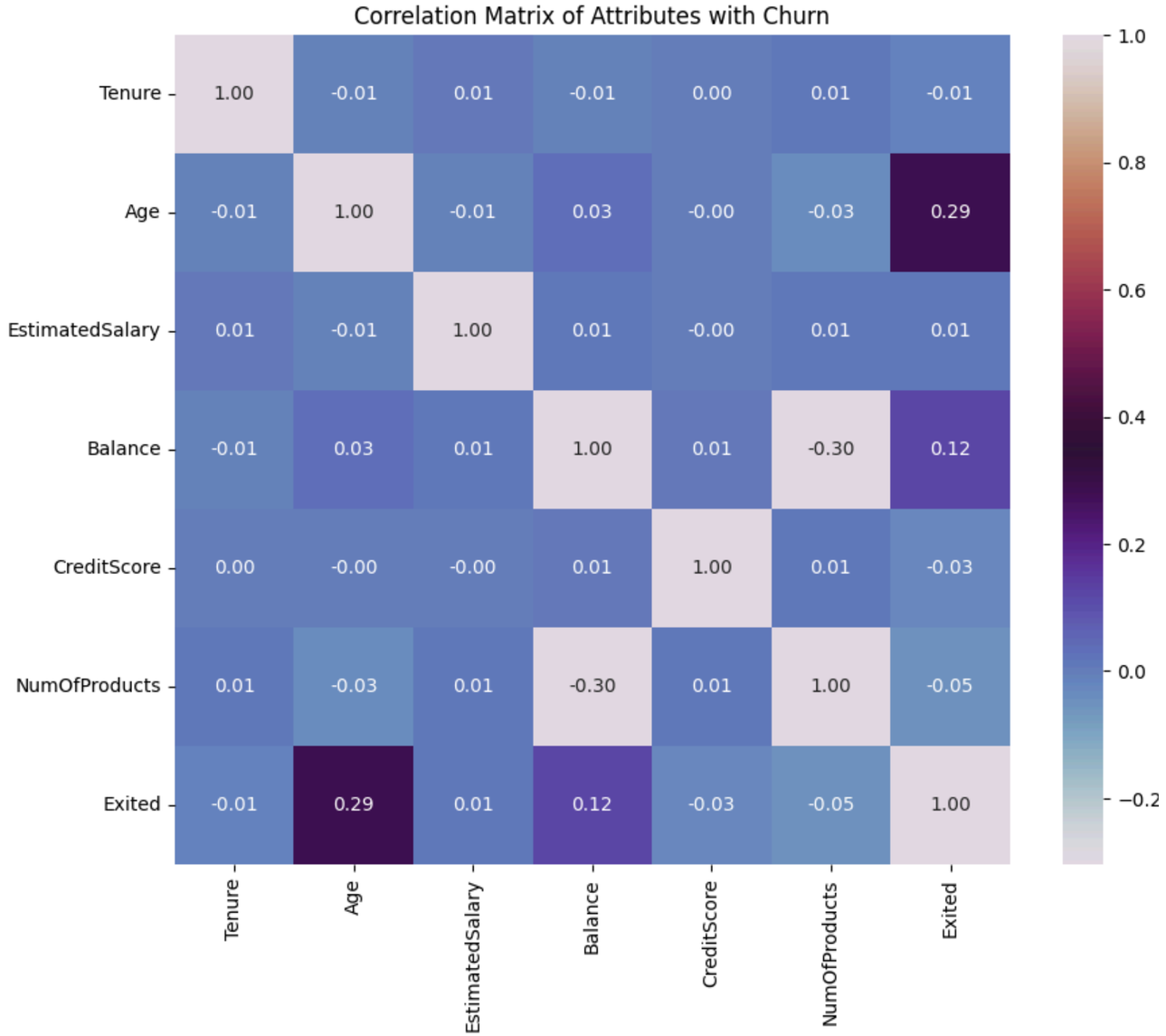
Q6

From Q5, Analyze through a Chart a.) Customer in which specific region is more likely to be "At-Risk" b.) Customer of which specific age group is more likely to be "At-Risk" c.) How many of the "At-Risk" Customer do not have a credit card ?

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
tomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProd	HasCrCard	IsActiveMember	EstimatedSalary	Exited	CreditScoreSegment	BalanceSegment	CLV	CreditScoreGroup	AgeGrc
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- As we can see from the above data set given in Q5,the number of customers who are At-Risk of attrition without a credit card is **1**.

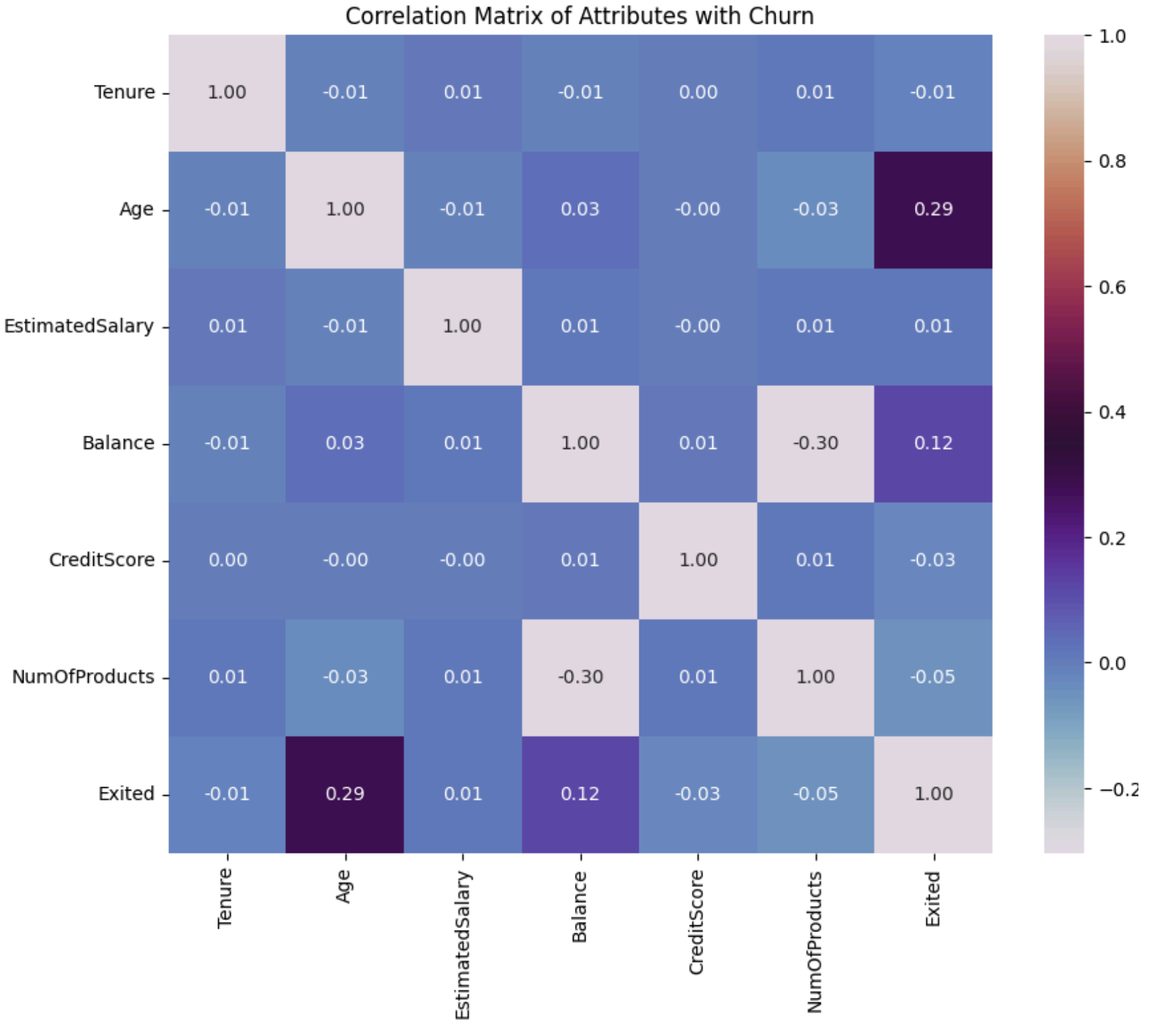
Q7 Discuss the correlation of the following attributes with Churn : Tenure, Age, Estimated Salary, Balance, Credit Score, NumOfProducts. Also give insights for each attribute.



- This is a correlation matrix showing relationships between various attributes and a churn indicator (labeled "Exited") in a dataset, likely for customer churn analysis.

Insights:-

- **Age** has the strongest positive correlation with churn **(0.29)**, suggesting that older customers are more likely to leave.
- **Balance** has a smaller positive correlation with churn **(0.12)**, implying that customers with higher balances may have a slight tendency to churn.



- **NumOfProducts** has a slight negative correlation **(-0.05)** with churn, suggesting that customers with more products are slightly less likely to churn, but this relationship is quite weak.
- **CreditScore** also shows a slight negative correlation **(-0.03)**, implying that a better credit score may be associated with lower churn, although the effect is minimal.

- Other Factors like **Tenure, EstimatedSalary, CreditScore, and NumOfProducts** have very weak or near-zero correlations with churn, indicating they have limited direct impact on customer churn on their own.

Overall, Age and Balance are the most relevant individual indicators of churn in this dataset

Q8

Estimate the Customer Lifetime Value (CLV = Tenure * Estimated Salary) and calculate the total value still with the bank (not churned). Analyze using charts : a.) Which region contribute most to the bank's revenue for non churned customers. b.) Does higher credit scores lead to higher CLV. c.) Visualize how the CLV is distributed among different age categories and geographical regions using Slicers

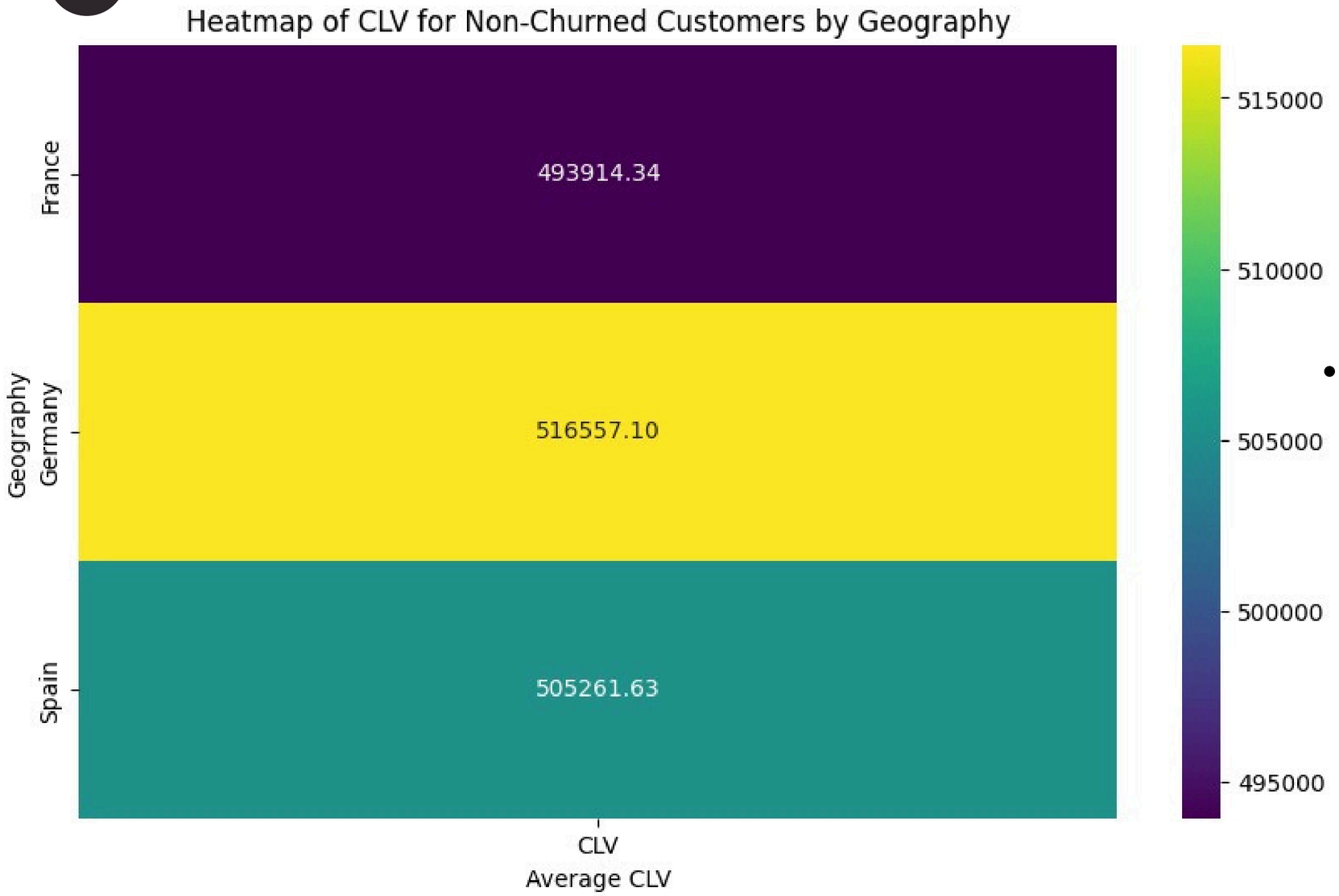
- The CLV for each costumer was calculated using the formula given in the question along with integrating it with the python libraries.
- Using CLV the total value still with the bank (not churned) was calculated by using the formula :

$$\text{Total value still with the bank} = \sum \text{CLV (non-churned}$$

- Total value still with the bank (non-churned) : **3994840162.1099997**
- **(Three billion, nine hundred ninety-four million, eight hundred forty thousand, one hundred sixty-two)**

Q8

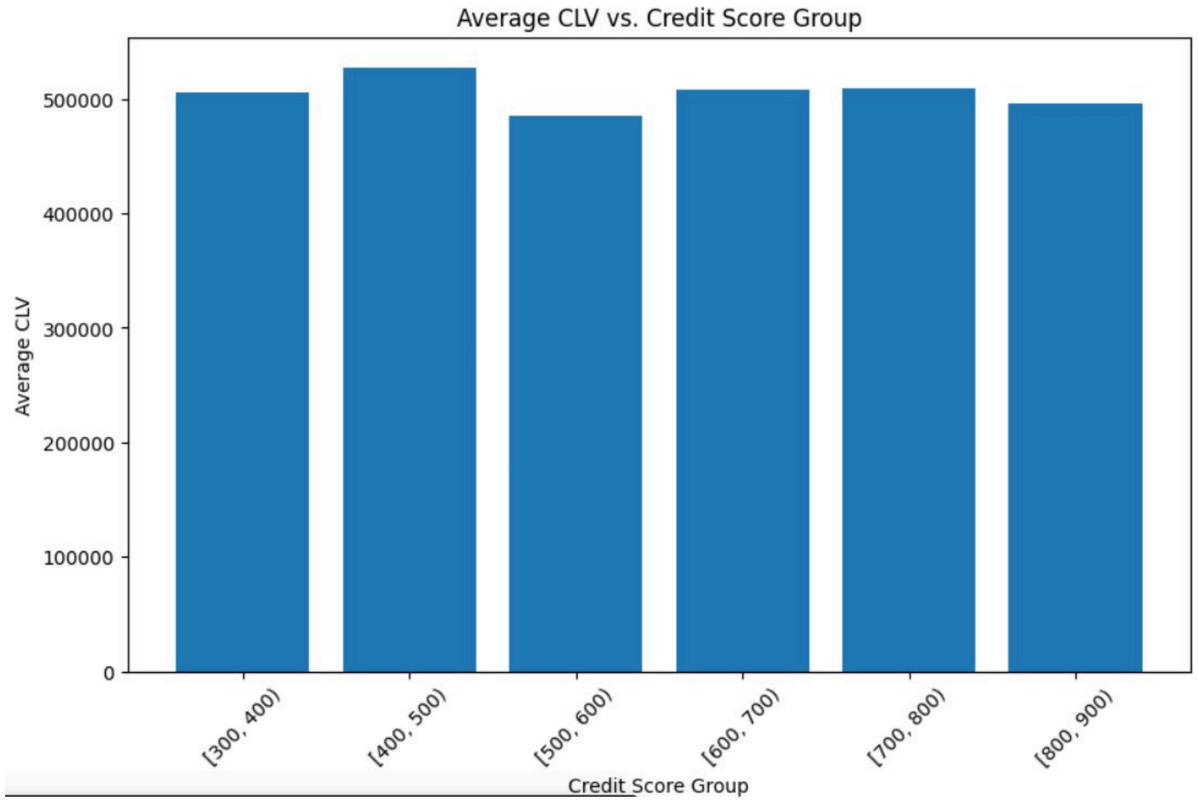
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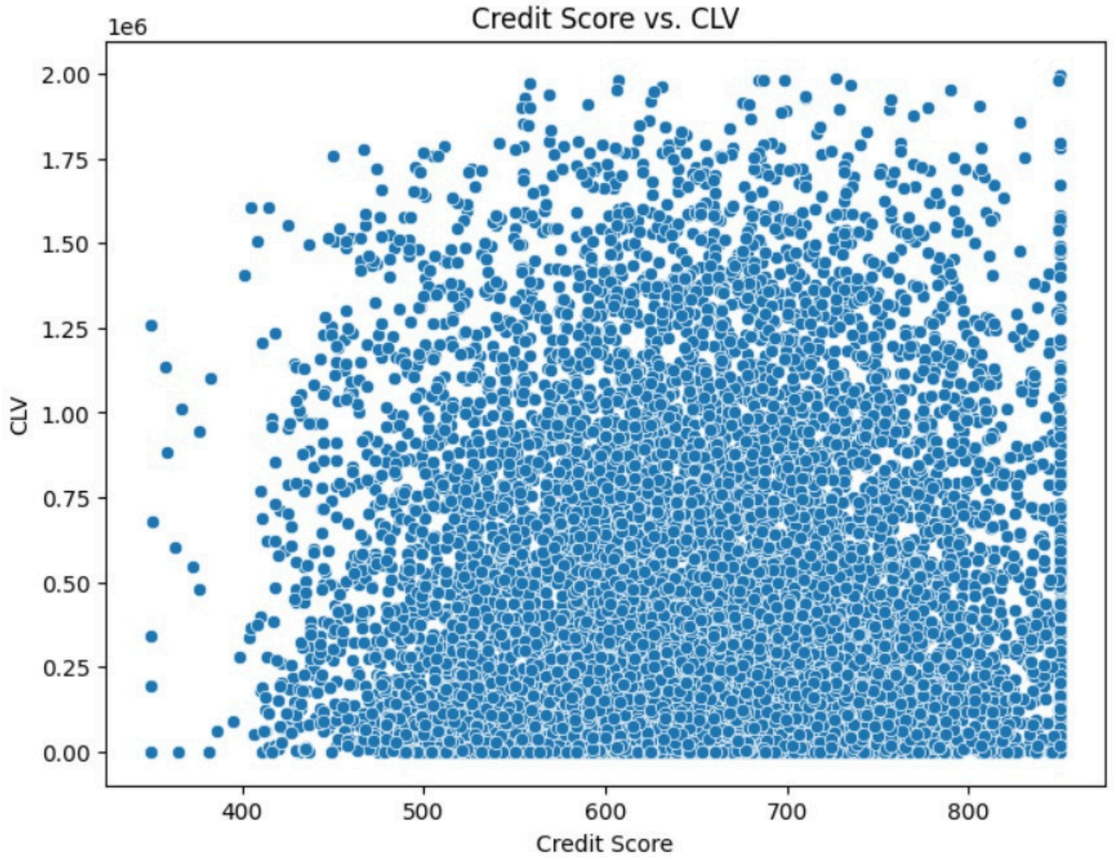


- Upon making a heat curve analysis is was observed that **Germany** has the highest number of non-churned costumers.

Q8

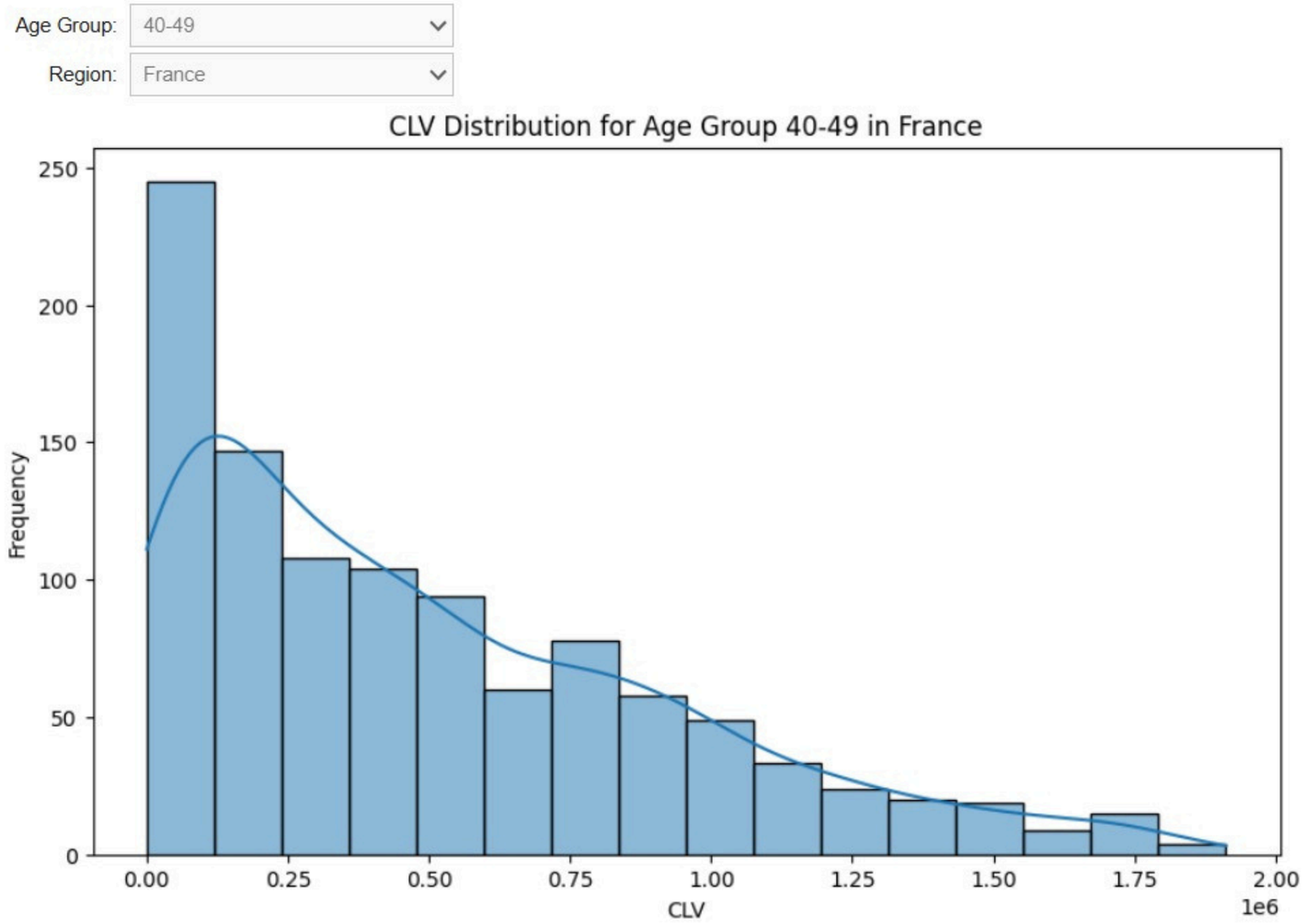
b





The average CLV value is almost uniform for all the credit score groups thus we can conclude that there is no direct relation between Credit score and CLV. Any relation found between the two variables for a particular data point can be considered coincidence.

Q8 C



- **X-axis (CLV):** The CLV is represented in millions. It likely represents the estimated revenue each customer is expected to generate for the business over their lifetime. The CLV ranges from 0 to 2 million in this chart.
- **Y-axis (Frequency):** This indicates the number of customers in each CLV range. The highest frequency is on the lower end of the CLV scale, indicating more customers with lower CLV values.
- **Distribution Shape:** The distribution has a right skew, with the highest frequency in the lowest CLV range (close to 0), which gradually decreases as CLV increases. This suggests that, for the 40-49 age group in France, most customers have a relatively low CLV, with only a few reaching higher CLV values.

- The smooth line over the bars represents a probability density function, showing the likelihood of CLV values within this range. The peak on the left side confirms that lower CLVs are more common for this demographic.

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THANK YOU

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